

Problem

Use convolution to find $v_o(t)$ in the circuit of Fig. 15.25(a) when the excitation is the signal shown in Fig. 15.25(b). To show how powerful working in the s -domain is, verify your answer by performing the equivalent operation in the s -domain.

Figure 15.25(a):

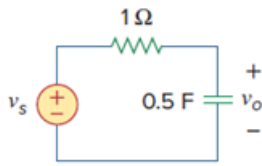
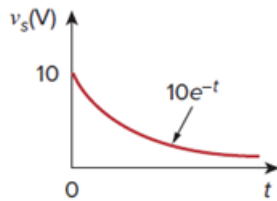


Figure 15.25(b):



Step-by-step solution

Step 1 of 7

Given input voltage is

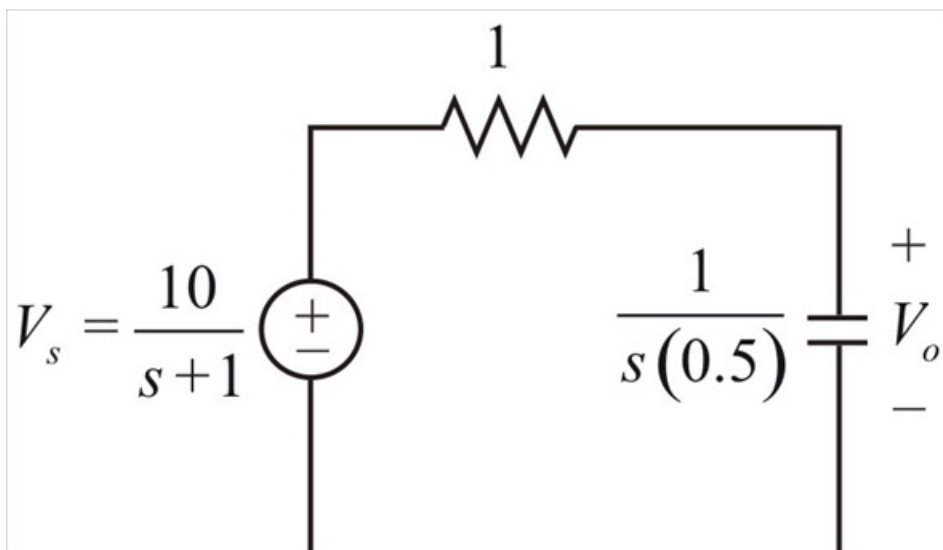
$$v_s(t) = 10e^{-t}u(t) \text{ V}$$

In s – domain

$$V_s(s) = \frac{10}{s+1} \text{ V}$$

The circuit in s – domain is

Step 2 of 7



Step 3 of 7

Using the voltage division rule, we get

$$V_o = \frac{\frac{2}{s}}{1 + \frac{2}{s}} V_s$$

$$V_o = \frac{2}{s+2} V_s$$

Hence

$$\frac{V_o}{V_s} = H(s)$$

$$\therefore H(s) = \frac{2}{s+2}$$

The inverse transform of impulse response is

$$h(t) = 2e^{-2t}u(t)$$

Step 4 of 7

Convoluting the impulse response and input voltage gives the output voltage

$$v_o(t) = h(t) * v_s(t)$$

$$v_o(t) = \int_{-\infty}^{\infty} h(\lambda) v_s(t-\lambda) d\lambda$$

$$v_o(t) = \int_{-\infty}^{\infty} 2e^{-2\lambda} u(\lambda) 10e^{-(t-\lambda)} u(t-\lambda) d\lambda$$

$$v_o(t) = 20e^{-t} \int_0^{\infty} e^{-2\lambda+\lambda} u(t-\lambda) d\lambda$$

$$v_o(t) = 20e^{-t} \int_0^t e^{-\lambda} d\lambda$$

$$v_o(t) = 20e^{-t} \left[\frac{e^{-\lambda}}{-1} \right]_0^t$$

$$v_o(t) = -20e^{-t} [e^{-t} - 1]$$

$$v_o(t) = -20[e^{-2t} - e^{-t}]$$

$$\therefore v_o(t) = 20(e^{-t} - e^{-2t}) \text{ V}$$

Step 5 of 7

To verify the output voltage by performing the equivalent operation in the s – domain

$$V_o(s) = \frac{2}{s+2} V_s(s)$$

We know that

$$V_s(s) = \frac{10}{s+1}$$

$$V_o(s) = \left(\frac{2}{s+2} \right) \left(\frac{10}{s+1} \right)$$

$$V_o(s) = \frac{20}{(s+1)(s+2)}$$

Using partial fraction expansion

$$\frac{20}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} \text{ ----- (1)}$$

Step 6 of 7

By multiplying $(s+1)(s+2)$ on both sides

$$20 = A(s+2) + B(s+1)$$

Consider $s = -1$

$$A = 20$$

Consider $s = -2$

$$(-1)B = 20$$

$$B = -20$$

Substitute values of $A = 20$ and $B = -20$ in equation (1)

$$V_o(s) = \frac{20}{s+1} - \frac{20}{s+2}$$

Taking inverse transform on both sides, we obtain the output voltage as

$$v_o(t) = 20e^{-t} - 20e^{-2t}$$

$$v_o(t) = 20(e^{-t} - e^{-2t}) \text{ V}$$

Step 7 of 7

Thus, verified